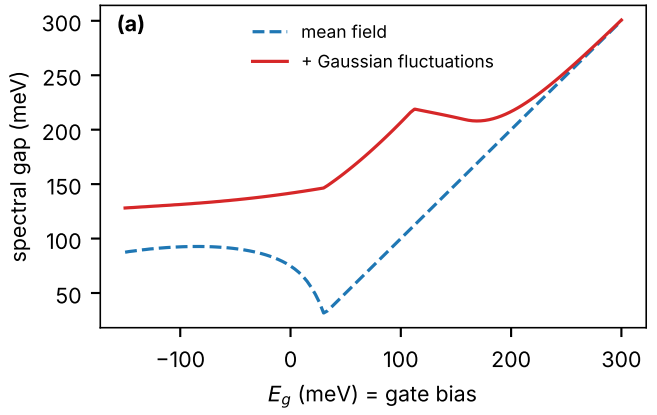
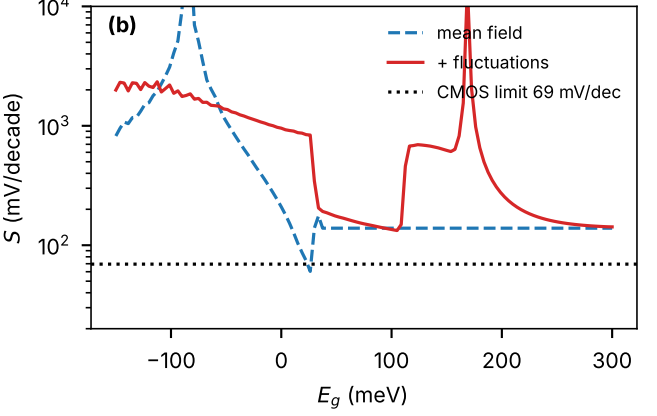

(a)
mean field
+ Gaussian fluctuations
spectral gap (meV)
300
250
200
150
100
50
−100
0
100
200
300
E_g (meV) = gate bias


(b)
mean field
+ fluctuations
CMOS limit 69 mV/dec
S (mV/decade)
10^4
10^3
10^2
−100
0
100
200
300
E_g (meV)